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PROJECT:	
SERIAL NUMBER:	

COMMERCIAL KITCHEN HOOD MANUAL

INSTALLATION, OPERATION AND MAINTENANCE FOR THE
URBAN ENVIRONMENTAL OPTI SERIES COMMERCIAL KITCHEN VENTILATOR SYSTEM

OPTI 100, 200, 300...L, R and LR MODELS
OPTI 1500, 1870, 2000, 2500 MODULES

Please read and save these instructions for future reference. Read carefully before attempting to assemble, install, operate or maintain the described product. Protect yourself and others by observing all safety information. Please contact your Urban Environmental Ltd. (UE) representative should you have any questions regarding the information contained in this document.

THIS DOCUMENT IS TO BE USED FOR INSTALLATION AND IS TO BE TRANSFERRED TO THE PRODUCT OWNER TO ACCOMPANY THE PRODUCT ON LOCATION. THE INSTALLERS AND THE END USER SHOULD BE FULLY AWARE OF ITS CONTENT.

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1. GENERAL INFORMATION

Please make sure that you have fully read the UE Terms and Conditions of Sales which have been sent to you together with your quote. If you have not received the Terms and Conditions of Sale, please contact your UE Agent.

1.1 General Safety Information

- Only qualified personnel should install or repair this kitchen exhaust hood.
- Personnel should have a clear understanding of these instructions and all applicable local and national building and fire codes.
- Personnel should be aware of general safety precautions.
- Building Code guidelines should be followed for hanging and supporting the hood.

1.2 Product Performance

UE can guarantee product performance if the equipment provided is installed, used, cleaned and maintained as per UE specifications and requirements. The use of third party parts within the UE kitchen exhaust hood, such as spare filters and cells, and/or inappropriate cleaning regimes will affect product performance and may be cause to void the product warranty.

It is up to the product Owner / End user to guarantee and be able to prove that the product has been used and cleaned as per the UE specifications and requirements. Should this not be proven, UE cannot be held responsible for any subsequent issues that may arise.

UE selects kitchen exhaust hoods and equipment in good faith and according to the information provided by others. The UE products are to be integrated into the overall mechanical design as per the UE defined specifications. In supplying equipment, UE cannot be held accountable for the overall performance of the mechanical system.

UE cannot be held accountable for product performance issues (for example excessive discharge of smoke, grease and odour) arising from the performance of the overall mechanical system including: duct design, fan selection, discharge point location, unknown cooking methods, and applied airflows other than the values specified by UE.

Filtration and odour mitigation products are specified based upon in-depth industry experience and existing equipment efficiency tests.

The overall performance of the specified product can only be guaranteed if a preliminary assessment is carried out by a suitable consultant, taking into account cooking types and regimes, discharge point sensitivity, airflow amounts and the overall mechanical system design. This is notably the case with regards to odour discharge. Otherwise, UE underlines that additional filtration and odour mitigation units may be required should there be any issues with the kitchen exhaust discharge during operations.

The UE OPTI Series hoods have been designed and certified according to various international standards. Contact your rep for details. The product complies with Australian and New Zealand standard AS 1668.2-2012, AS 1530 the NZ Building Code G.

The UE exhaust hood product specification and drawing sheet is to be checked and approved by the Client, prior to start of manufacturing or delivery. It is the responsibility of the Client to ensure that the UE hood is suitable for its proposed application within the guidelines shown on the specification drawing, that all services to and from the hood are as required and that site access and installation constraints have been taken into account.

In order to assist installation, UE provides the following installation, operation and maintenance guidelines showing typical installation cases. It is up to the Client to fully review the requirements of the product which may have deviations to the guideline drawings and for the specific installation. It is the responsibility of the Client to contact the local UE representative

should they require any further information regarding performance, installation, commissioning and maintenance of the product. All UE products are to be installed following the locally applicable code and standards.

The commissioning of the product is to be carried out by a recognised mechanical contractor. All airflows should be as provided as per the relevant kitchen exhaust hood specification sheets and related documentation.

UE disclaims liability or responsibility for damage from fires and explosions regardless of origin. It is the customer's responsibility to ensure that appropriate fire protection systems or explosion venting are installed following the requirements of local applicable codes and standards.

1.3 Delivery and Handover

Unless otherwise stated, UE delivers kitchen exhaust hoods to the nearest site access point. Unloading of the truck is the responsibility of the Client.

Unless otherwise stated, delivery of the hood will commence once the full invoiced amount has been received by UE.

Delivery of the hood will be to the address as indicated by the Client and by means of the agreed logistics.

Unless otherwise requested, the delivery will be insured to the closest suitable access point, at which point the handover of the hood is completed. The Client is responsible for unloading of the hood from the delivery vehicle, which may require specific lifting equipment.

UE will inform you on manufacturing progress approximately one week prior to the planned delivery of the kitchen exhaust hoods. At this time, UE will request information regarding

- Exact delivery location
- Site access constraints including limited delivery times
- Site contacts
- Any additional information that may affect delivery of the equipment.

UE will also confirm the delivery method and inform the Site Manager with regards to the unloading of the trucks. A detailed Packing List will then be supplied summarising the overall delivery conditions.

Alternatively, the Client can organise to have the product picked up at a UE warehouse or manufacturing site, at which point handover is completed.

If damage occurs during shipping, the Client is to notify UE as of the delivery date. Photographic evidence should be provided so that a claim can be filed.

Unless otherwise mentioned, cost of delivery including freight and any additional handling costs (including crating), are not included in the UE quoted price and will need to be organised once a delivery address is provided.

The kitchen exhaust hood will be delivered at the agreed delivery date and following on from UE quality control and testing on the equipment.

UE reserves the right to charge for storage of the hood after 1 (one) week of the agreed delivery date, should the Client request that the hood delivery be postponed or if the conditions for delivery have not been fulfilled. Should hoods be stored on factory floor transport costs to UE warehousing may be charged in order to reduce ongoing storage costs.

1.4 Receiving

Upon receiving the product, check to ensure all items are accounted for by referencing the delivery receipt or packing list.

Inspect each crate or carton for shipping damage before accepting delivery. Alert the carrier/driver of any damage detected. Please make a notation of damage (or shortage of items) on the delivery receipt/packing list which is to be countersigned by the delivering driver.

If the equipment is damaged, immediately contact the UE Agent. Any physical damage to the unit after acceptance is not the responsibility of UE.

1.5 Unpacking

Verify that all required parts and the correct quantity of each item have been received.

If any items are missing, report shortages to the UE Agent to arrange for obtaining missing parts. Sometimes it is not possible that all items for the unit be shipped together due to availability of transportation and truck space.

Confirmation of shipment(s) must be limited to only items on the packing list. Filters are shipped in their original packaging and are not installed in the kitchen exhaust hood. Do not remove factory packaging or install filters until just prior to commissioning.

1.6 Handling

Units are to be rigged and moved by the lifting points provided or using the pallet when a forklift is used.

Handle in such a manner as to keep from scratching or denting. Damaged finish may reduce the ability of the unit to resist corrosion.

1.7 Storage

Units are packaged with protection against normal damage during shipment. But the items are large and heavy and cannot consistently withstand being pushed over or dropped. If the unit cannot be installed and operated immediately, precautions need to be taken to prevent deterioration of the unit during storage.

The Client assumes responsibility for the unit and accessories while in storage. UE will not be responsible for damage during storage. Shipping liabilities depend on prior arrangements.

The ideal environment for storage of the hood and accessories is indoors, above grade in a clean, dry atmosphere that is sealed from the elements. While in storage, inspect equipment routinely. If any moisture, dirt or other accumulations are found on the hood or any of the parts, the source should be located and eliminated.

2. COMPONENTS

2.1 General Specifications

The following schematics are for standard non customised versions to the complete UE OPTI kitchen exhaust hood and detail an OPTI1500LR model with a single centre module. For longer configurations and the exact specifications to the purchased kitchen exhaust hood, please refer to the project specification sheets.

UE Centre Modules are supplied standard in the following lengths 1500mm,2000mm and 2500mm. Modules are designed to be attached to each other to form large exhaust hoods. Modules will be supplied generally with end pieces, left (L, OPTI 0003) and right (R, OPTI 0004) to be fixed to each end. End modules each add 100mm to the total length

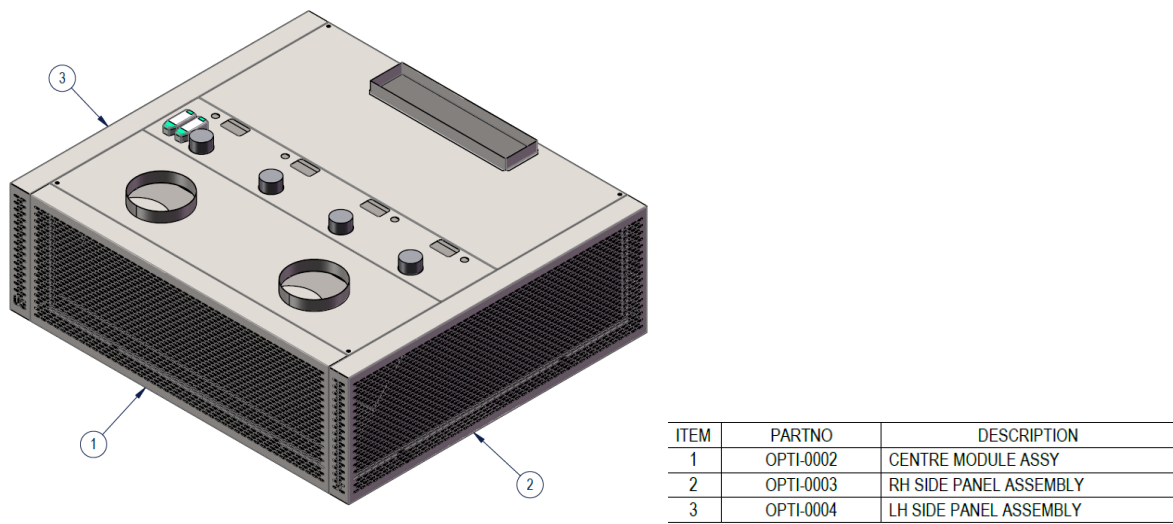


Figure 1

2.2 Parts List

Each module comes with the following independent parts. The quantity of parts will vary depending on your configuration. This list is based on a single OPTI 1500 LR module as pictured on the previous page and throughout. Some parts are provided loose to aid in the assembly and installation process.

Parts listed below are referenced in the following pages in this manual. If the supplier is not listed or available contact UE for supply.

ITEM	PARTNO	DESCRIPTION	SUPPLIER	QTY
1	OPTI-0006	FRONT PANEL STAND-OFF ASSY		6
2	OPTI-1001	CENTRE MODULE FABRICATION		1
3	OPTI-2001A	FRONT AND SIDE PANEL		1
4	OPTI-2007C	REMOVABLE PANEL		1
5	OPTI-2008E	DAMPER INFILL		2
6	OPTI-2009A	LIGHT COVER PANEL		1
7	OPTI-2009B	LIGHT COVER GLASS		1
8	OPTI-2009C	GREASE COLLECTION CUP		1
9	BHCS-SS304-M05X010	BUTTON HEAD CAP SCREW - SS304 - M05X010	FASTENER	1
10	BHCS-SS304-M05X020	BUTTON HEAD CAP SCREW - SS304 - M05X020	FASTENER	21
11	CSKS-SS304-M05X030	COUNTERSUNK SOCKET SCREW - SS304 - M05X030	FASTENER	9
12	NUTN-SS304-M05	NYLOCK NUT - SS304 - M05	FASTENER	20
13	WASF-SS304-M05x15x1.2	WASHER LARGE - SS304 - M05x15x1.2	FASTENER	40
14	JLCKKH082 - 495X254	HONEYCOMB FILTER		3
15	JLKH-161	BAFFLE FILTER 370W x 495H x 45D		4
16	ICDLTWINR12W3K-BCH	12W TWIN LED DOWNLIGHT PACK INCL DRIVER	GEO LED	2
17	4860 - Ø55	VENTILATOR FOR Ø55 HOLE	POMMIER	2

2.3 Tools Required

3mm allen key

4mm allen key

8mm wrench socket

19mm wrench socket

Food grade heat tolerant silicon sealant

Caulking gun

Heavy lifting equipment

Height access equipment

Specific hanging hardware and equipment

3. ASSEMBLY

3.1 Overview

UE hoods are provided packed and pre-assembled. Delivery is normally roadside so and packaging is not normally waterproof. make sure you're familiar with 1. GENERAL INFORMATION section of this manual and get excited. The following section provides a step by step guide to hanging but be aware every location may have special requirements. With the correct lifting equipment and a clean prepared site two people should be able to have a normal size hood fully installed within a few hours.

3.2 Guide

The OPTI hood is provided in easily managed sections to allow for access. Final assembly needs to happen on site. Below are the assemblies ready for hanging. The centre module has panels and filters removed so the internals can be accessed.

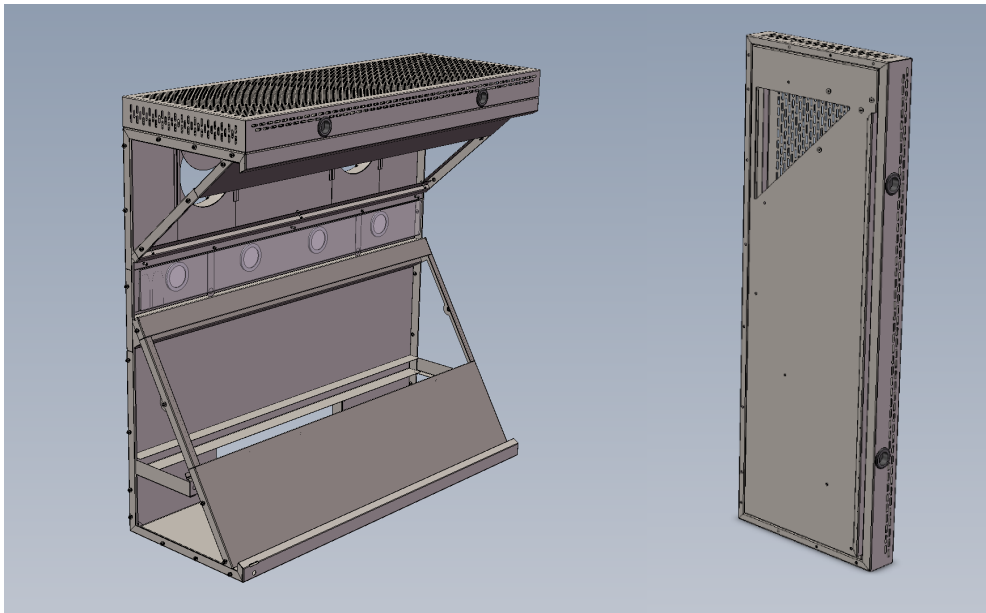


Figure 2

ITEM	PARTNO	DESCRIPTION
1	OPTI-0002	CENTRE MODULE ASSY
2	OPTI-0003	RH SIDE PANEL ASSEMBLY
3	OPTI-0004	LH SIDE PANEL ASSEMBLY

Warning: These schematics should only be used for reference to specific components and should not be used to plan the full installation of the kitchen exhaust hoods. The schematics are in no way to be used as a replacement to detailed kitchen exhaust specification sheets and direct assessment on site.

Note: Longer hoods have multiple centre modules (OPTI 0002). See examples of OPTI 2 x 1500 and 3 x1500 set up below.



- Unpack hoods as per and layout parts
- OPTI 0002 modules should be set on their backs, as pictured in the area where they are to be installed and bolted together using an allen key, crescent and the fasteners provided.
- Bolt ends to hoods and seal joints make sure that all bolt fixing points as seen in SECTION C-C are fitted securely. Nut and washer to the one side as in DETAIL A and B in [Figure 3](#) and locations in [Figure 4](#). There are 19 bolts required per module join (M5X20MM). See more detail in section [3.5 Joining Modules](#).
- **It is strongly recommended to attach universal structural strut to the top of the hood greater than one module long.** This is set up for longitudinal fitting at the front and back of the hood using all bolt holes and 38mm wide channel. This provides strength and additional alignment across the length of the hood. Channel can also be fit transversely if required.
- Skip this point if the front perforated panel is already attached. These panels (2001A) can be fitted before or after lifting. See detail D in [Figure 4](#) and also [9. OPERATION AND MAINTENANCE](#). The front panel attaches at 6 points with assembly (0006) including 6 x bosses and 12 x bolts. Bosses are fitted tight to the hood section through the makeup air access panel (2007C). Front facing screws are left loose for keyhole fittings and final tightening is made with an Allen key through the front perforations.
- Before lifting check your safety requirements. Install additional damper plates or part OPTI 2008E, if you anticipate they will be needed, leave out filters, light cover (2009A,B) access panel (2007C) and grease collection cup (2009C)
- Lights are colour changeable by a switch on top of each spot. set to desired colour.
- Set hood flat/horizontal on the floor or on your lifting equipment. Attach the rigging frame and equipment to lift the assembled hood evenly into position. Be careful to make sure that the full hood is always supported evenly. Support is required at every join and on the ends.
- Level the hood to the recommended height. Most often 2000mm AFFL but some local authorities may require different heights. Often heavy machinery is needed to do this safely. See hanging [3.6 Preparations for Hanging](#) section for details.
- Fix the hood to walls with coach bolts or threaded rod to support above. Use lock tight and double locking nuts.
- Once hood is hung fit access panel (2007C) and grease collection cup (2009C)

- Once ducting is connected, fresh air can be adjusted. See detail in Figure 6 . If it is possible you can do this before fitting the light cover (2009A,B).
- See section [4.3 Spigots](#) for details on connecting ducting to the hood.
- Exhaust air can be adjusted on some models with a similar mechanism as in [Figure 6](#). If it is possible you can do this before fitting the light cover (2009A,B). If necessary exhaust flow can be further adjusted by fitting sliding plates. Part labelled 5 in Figure 3 can be used (OPTI 2008E).
- LED lighting comes with a power plug and 2m cable. This can be connected to the room lighting circuit by a GPO mounted above the hood at the left end of each module or plugs can be removed and hard wired by an electrician.
- Ducting is normally formed to fit inside the exhaust spigot and outside the make up air spigots. See [4.3 Spigots](#). The connection should be fully sealed.
- Filters and remaining parts can be installed if desired, or stored until other construction on site is complete.

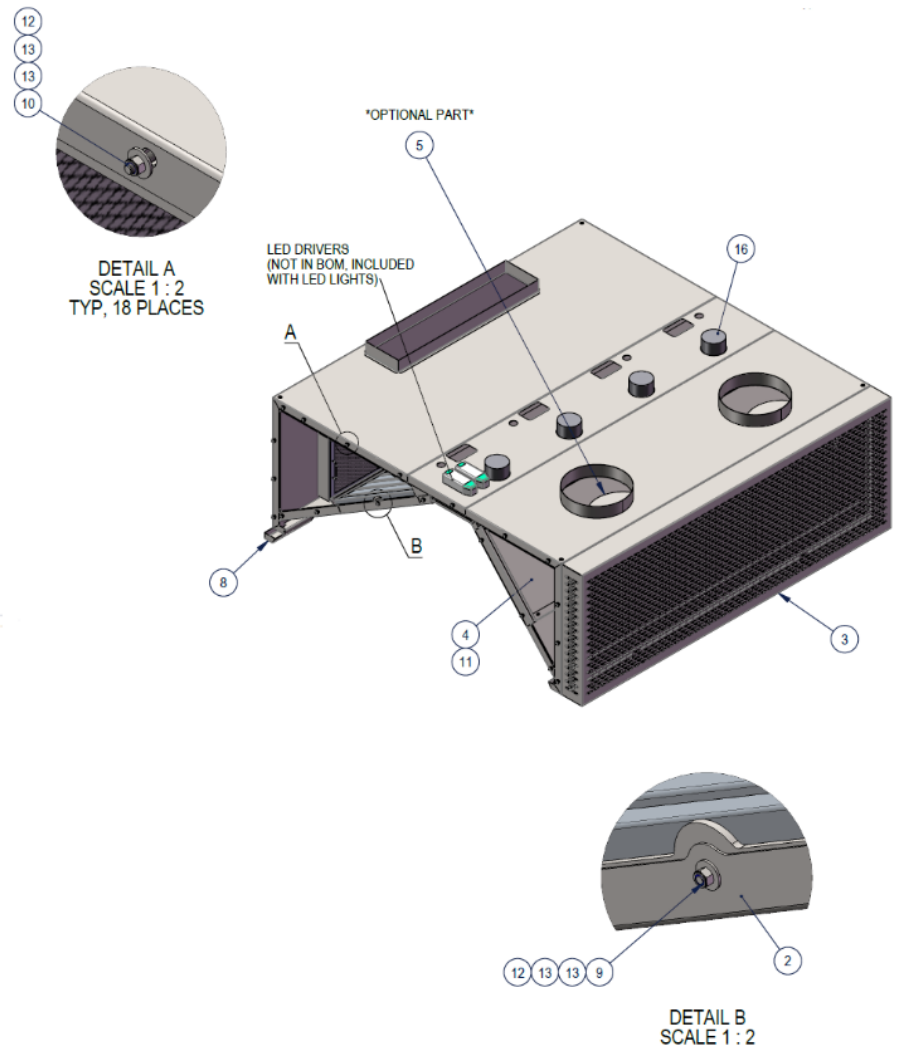


Figure 3

3.5 Joining Modules

- Peel back PE wrap from steel surfaces that will be between connecting sections.
- Attach ends to the sections of the hood that are to form the ends of the hoods. Apply 2mm silicone beads between joints before bringing parts together.
- Move the sections to be joined close together and add seal joining faces as needed.
- Whilst still sitting on the ground on their back pull the hoods sections together first fixing hand tight two bolts at the top corners followed by the two bottom corners.
- Tighten these four corners until the assembly is fully closed.
- Install the remaining bolts as in DETAIL A and B in Figure 3 and make sure they are all fully tight.
- Follow this up by re-tightening the four corners and remaining bolts.
- All fitting holes are to be equipped with the required bolts. There is an additional key hole bolt hole front and back in case further tightening is needed.
- Modules need to be sealed together using a non-acetic silicone sealant to stop grease or water leaks at the joints.

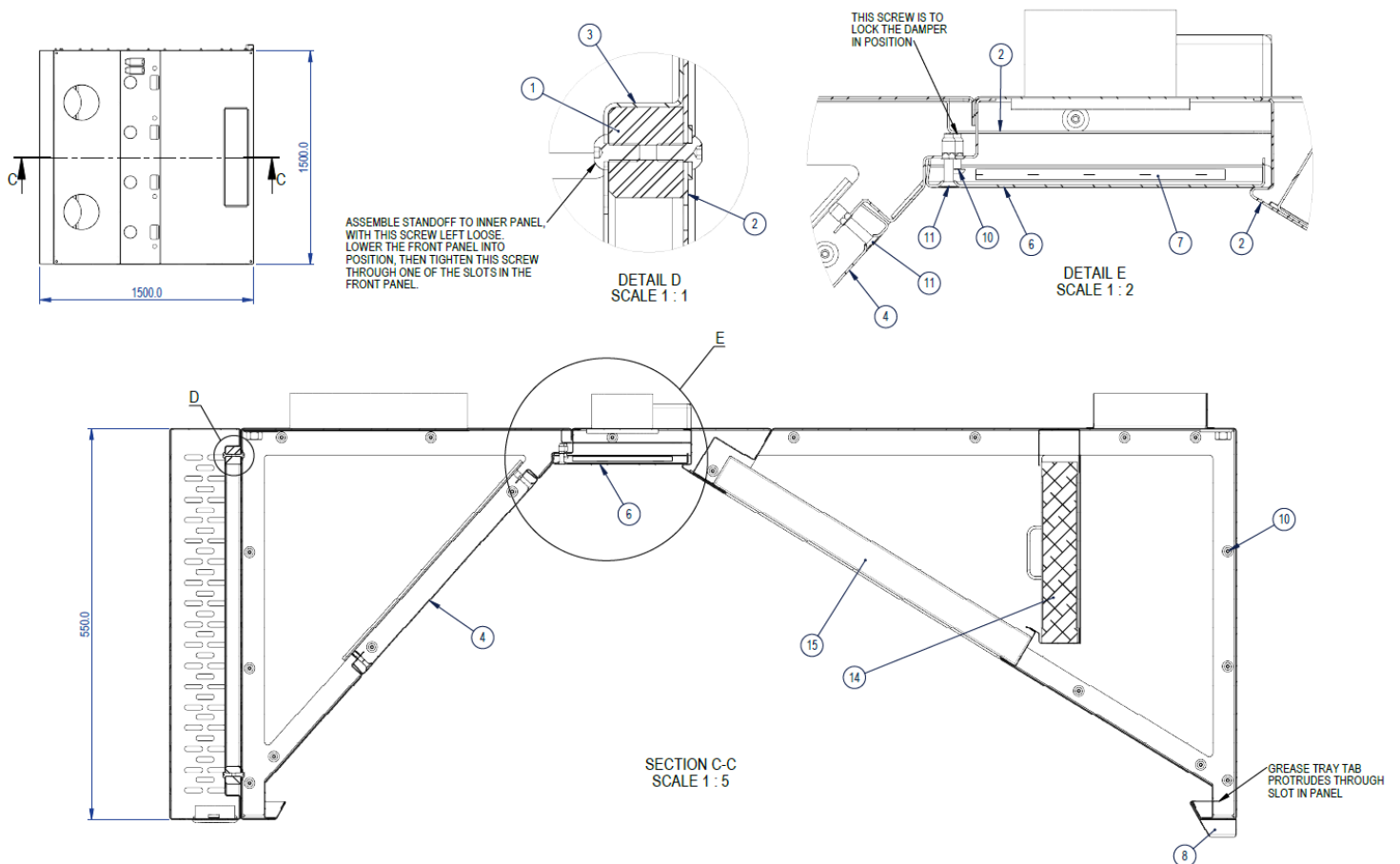


Figure 4

3.6 Preparations for Hanging

Hanging a large kitchen hood can be a formidable and dangerous thing. They are heavy and there are significant health and safety risks including loss of life. Before hanging your hood, you will want to have the overhead hanging system ready to go. Hoods can be fitted directly to the wall, but some installations will require seismic supports from above. This can be done with commonly available threaded rod, chain, wire and strut systems.

Each hood section is fitted with 4 welded fittings where stainless steel M12 bolts can be fitted as below in Figure 5. The fitting points can be rated at 19kN Proof Load, 33kN Tensile Load, CLASS 4.6 TO AS 1110 each.

An additional M12 nut is to be added below the one shown in Figure 5 and tightened creating a vibration resistant locking nut fixing.

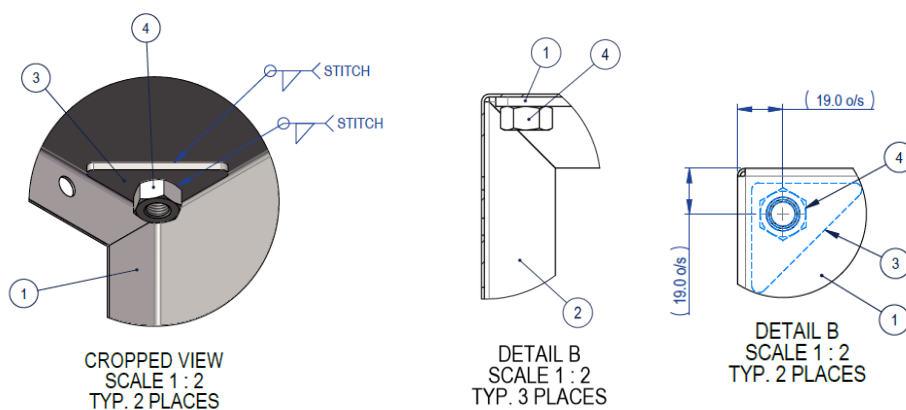


Figure 5

ITEM	PARTNO	DESCRIPTION	MATERIAL	FORM	SIZE
1	OPTI-2005A	REAR PANEL	AISI 304	SHEET	1.2
2	OPTI-2005B	REAR SPIGOT	AISI 304	SHEET	0.90
3	OPTI-2005D	LIFTING PT REINFORCEMENT	AISI 304	PLATE	4.0
4	NUTH-SS304-M12	HEX NUT - SS304 - M12			

Ducting may need to be installed above the hood before hanging the hood depending on your specific installation. See [4.3 Spigots](#) and [8. HOOD SPECIFICATION SHEET](#) to line up your ducting and make sure space is left for duct access and lighting transformers along with any other components to go on top of the hood and power connections.

Fire protection systems may need to be installed above the hood before hanging the hood depending on your specific installation see [8. FIRE PROTECTION](#). You may want to cut holes in the hood for this before hanging the hood.

For details on maneuvering the assembled hood to position see [4. HOISTING AND FITTING](#)

3.7 Final Assembly

After the hood is hung it is recommended to leave plastic steel protection on the outside surfaces of the hood for the full period of construction or as long as possible.

Make sure that the hood is true and level. Once this is complete do a final check and tighten over all of the fixing points.

Arrange dampers for commissioning. Normally dampers will be set all the way open so it is recommended to fix these firmly at this stage.

You can install the glass light cover if not already installed, but these will need to be removed for commissioning. It is recommended to leave these parts off along with the hood filters and store them safely together.

Silicon caulking can be added to joints and to the back wall where it meets the hood.

If a wall is present the hood should fit tight against a straight and plumb wall with fire protection.

NOTE: POSITION DAMPER SO THAT SCALLOP IS CONCENTRIC WITH AIR INLET HOLE IN TOP PANEL AND HANDLE PROTRUDES THROUGH SLOT IN OTHER PANEL AS SHOWN.
ARRANGE JOGGLE STRIPS SO THAT DAMPER CAN SLIDE FREELY, BUT THERE IS NOT TOO MUCH SIDEWAYS PLAY.
STITCH WELD JOGGLE STRIPS IN PLACE AND CHECK THAT DAMPER CAN STILL SLIDE FREELY.
TYPICAL, 2 PLACES

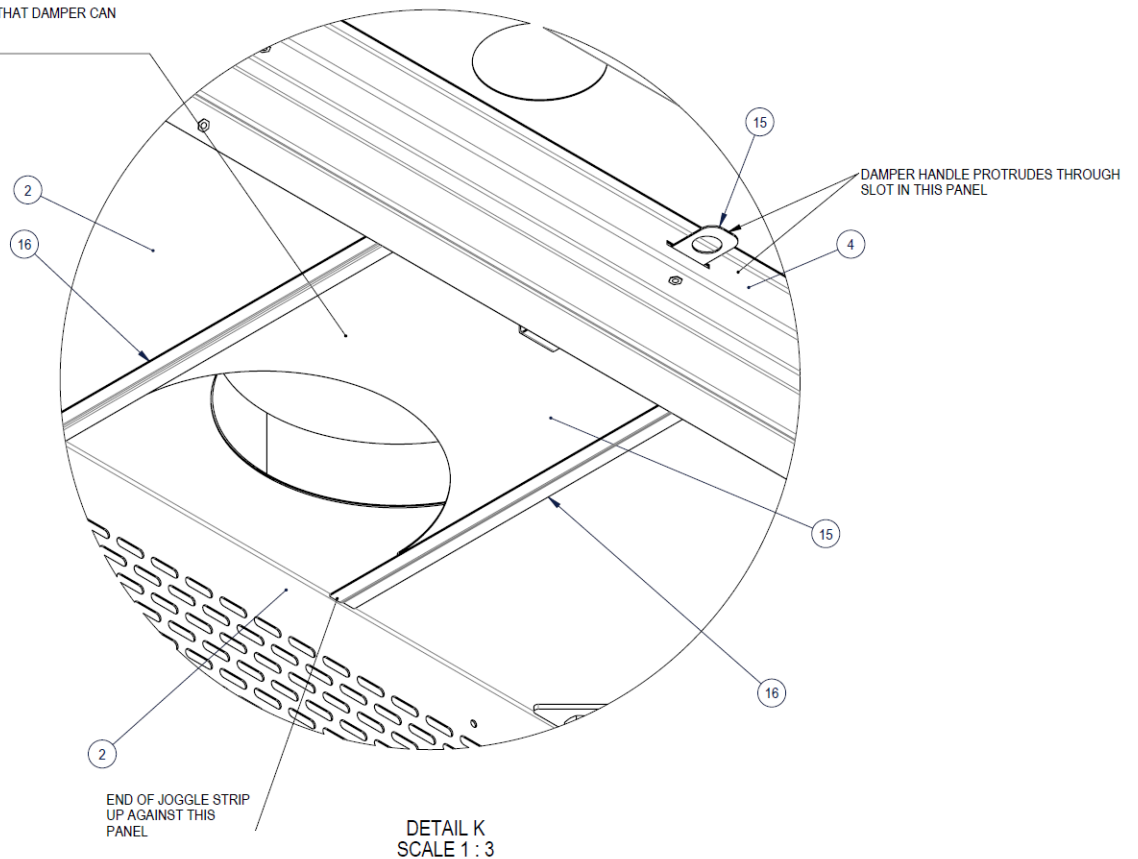


Figure 6

4. HOISTING AND FITTING

4.1 Prior to Installation

UE recommends that the hood be installed by a recognised mechanical contractor UE certified installers may be available..

Verify that there is enough space for the hood and lifting equipment in the operating position and enough clearance around the hood once it is in for components like fire system connections, hood lights, hood control components, etc.

Consider access for servicing the equipment and the different components when locating the hood. Note in particular that the front panel is designed to lift up and off requiring 15mm clearance above.

The kitchen exhaust hood specification sheet is found in [8. HOOD SPECIFICATION SHEET](#) of this document. The specification sheet will provide pertinent information required for hood installation including but not limited to:

- Hood weight
- Positions of exhaust and supply connection points.
- Equipment overhang requirements, notably side overhang.
- Specified exhaust and supply airflows applicable.
- Maximum exhaust and supply airflows applicable.
- Hood lighting electrical connection details.

Should any further information be required, please contact your UE agent.

It is the responsibility of the Client to ensure that the existing conditions allow for the nominated weight.

This installation guideline is limited to providing guidance to mounting an UE kitchen exhaust hood. It is up to the installer to fully review the requirements of the purchased hood which may have deviations to the guideline drawings.

The hood is to be hoisted and fitted according to applicable codes, regulations and standards.

4.2 Important Installation Information

It is recommended that the protective plastic sheeting remain on the hood until fully installed or the restaurant is opening to better protect the product from scratching and marking. Once removed, use stainless steel polish to clean the hood and/or remove marks or discoloration.

4.2.2 If the kitchen exhaust hood is provided in separate module parts to form a larger hood, the parts are to be bolted together prior to hanging and fixing the hood see [section 3.5](#).

4.2.5 To lift the hood into position, the following minimum requirements are recommended.

- No filters should be in the hood structure and the hood should be empty of any movable parts. Access panels and light covers are also removed.
- The hood needs to be supported evenly through the full length. It is recommended to fit universal type mechanical strut to the top of the hood running transversely across the front and back edges of the hood.

- For hoods over 3m in length and hoods that have been supplied in parts, it is recommended to position multiple genie lifts of adequate capacity so that the weight of the hood is well balanced. The lifts are to be operated simultaneously. Not doing this may deform the hood.

4.2.6 Once at the specified position, normally 2000mm affl. with 300mm overhang, see the [specification sheet](#) for details. The hood is then to be suspended from using threaded rods to the structure above either direct to the hood or to universal strut bolted to the hood— both methods using all fixing points.

4.2.8 The gutters running along the length of the hood are to be level or slightly falling to the left. The side gutters if not level are to tilt to the back slightly to enable grease and water to run towards the grease cup.

Warning: When mounting the hood or any components against the hood, never puncture or drill into the canopy without fully sealing the penetration. Incorrect cutting and drilling of the hood will void the hood listing and warranty.

4.2.10 Follow the kitchen exhaust hood specification sheets which refer back to AS1668.2-2012 for information on:

- Floor to underside of hood height.
- Side overhangs over the cooking line. The kitchen exhaust hood should be positioned so that the side overhangs follow the nominated overhang on the hood specification sheets.
- Front and rear (for island configuration) overhangs of the cooking equipment.

4.2.11 The kitchen exhaust hood needs to be positioned hard up against a wall. Bolts through the back wall of the hood are permitted but should be fully sealed and should consider possible heat transfer in the event of fire. Should, for any particular reason, there be a gap between the hood and the wall, this needs to be covered using stainless steel and fully sealed.

4.2.12 Once the hood has been fixed and positioned, filters can be installed. The access panel can be fitted. The light cover can remain off until commissioning.

4.3 Spigots

4.3.1 Typical exhaust spigot

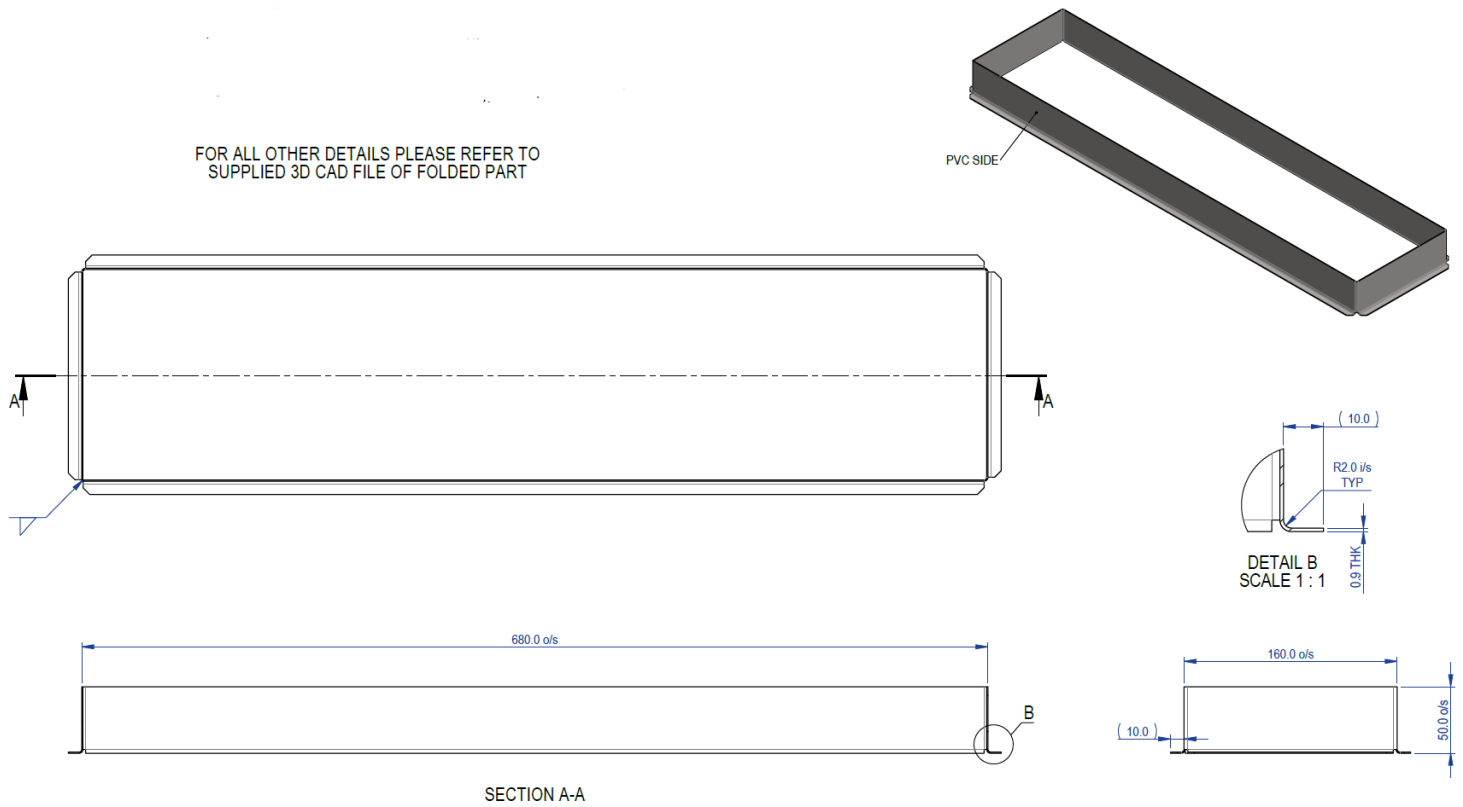


Figure 7

4.3.2 Typical air supply/make-up air spigot

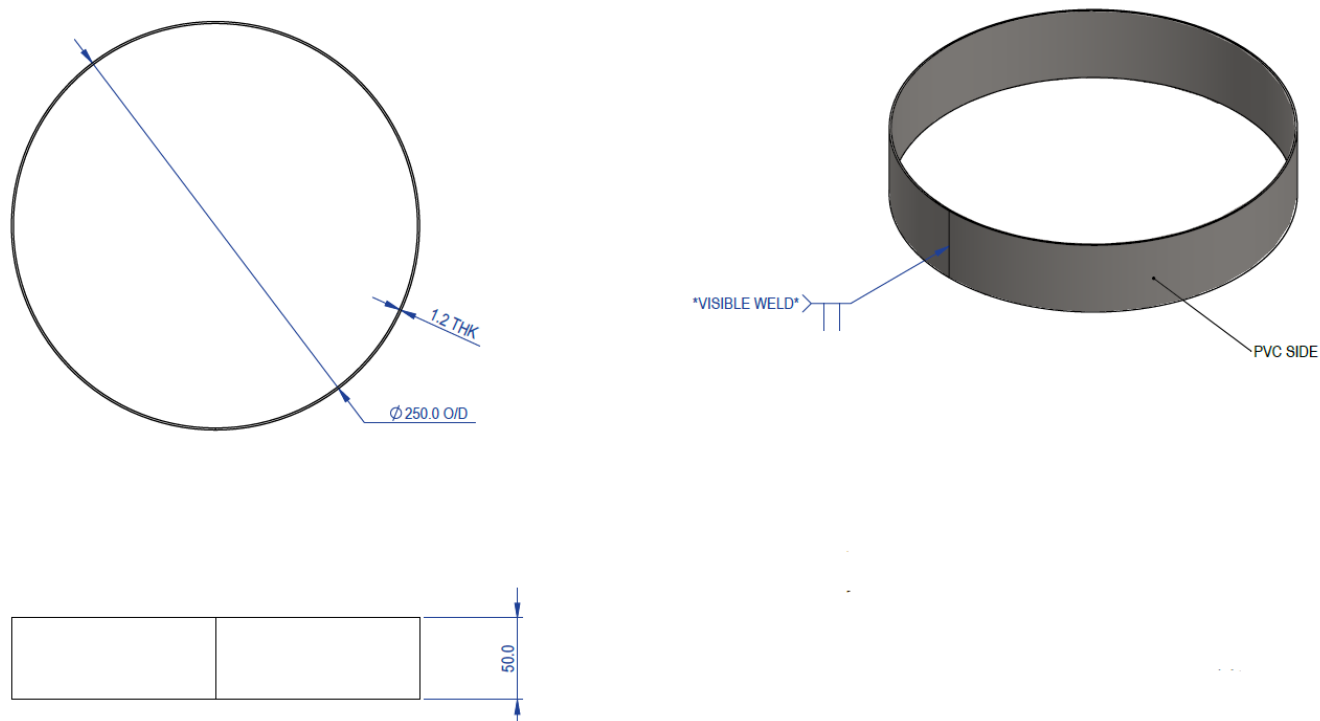


Figure 8

5. MECHANICAL CONNECTIONS

5.1 Connections to Exhaust Points

5.1.1 Exhaust ducting is to be connected to the kitchen exhaust hood at the position of the exhaust opening found on the hood specification sheet.

5.1.2 Each exhaust point is equipped with a sliding damper to allow for the airflow balancing. The access to the damper adjustment is behind the light cover or through the filter cavity and the internal hood exhaust plenum.

5.1.3 Details of standard hood connection spigots are shown in [Figure 7](#) and [Figure 8](#)

5.1.4 Exhaust ducting is to have an 'aircon' role formed flange fitted or be fixed to the inside of the spigot and properly sealed. Use duct of outside dimension 158x678mm for internal fitting.

5.2 Connections to Air Supply Points

5.2.1 Supply ducting or round flexible ducting is to be connected to the hood at the dedicated supply air points. The position of the supply air points can be found on the hood specification sheet.

5.2.2 Modules are provided with dedicated spigots outside diameter 250 mm equipped with dampers. Supply ducting is to be flanged, taped or fixed to the spigot and properly sealed.

5.2.3 The access to the supply spigot is done by removing the inner panel. Damper can be adjusted from behind the light cover.

6. ELECTRICAL CONNECTIONS

6.1 LED Light Connections

6.1.1 Power connection for LED lighting can be hard wired to the LED drives or depending on your country by 2 meter long three pin standard plugs. Connection is located on top of the kitchen exhaust hood at the left hand end of each module.

6.1.2 The number of connections depends upon the size of the kitchen exhaust hood and is detailed on your specification sheet including amp of the feed and wattage used. Normally one connection per hood section with a 10 amp feed..

6.1.3 Depending upon the requirements of the end user the hood lighting can either be powered ON with the same switch as the fan/filtration equipment, or preferably connected to the room lighting circuit or a separate switch.

6.2 Connection of Filtration and Other Equipment.

6.2.1 Filtration equipment may have some connections that need to be hard wired directly into the hood or the control box. Please check if this is included. This may include heat sensors and infrared sensors mounted in the hood.

6.2.2 Active hood systems with sensors and filtration often require commissioning. Contact your UE representative for help on this.

6.2.3 If mains connected filtration equipment is to be included it is to be powered on the same switch that controls exhaust fan operation. so that the filtration can not run if the fan is not on.

6.2.4 The number of connections depends upon the size of the kitchen exhaust hood and is detailed on your specification sheet including amp of the feed and wattage used.

6.2.3 If relays are included in control boxes they can be used to aid control and monitoring solutions. it is recommended to connect them. Details will be included in a separate manual for your filtration equipment.

7. AIR BALANCING

7.1 General

- 7.1.1 Both the maximum exhaust and supply airflows applicable to each hood and each module are shown on the hood specification sheets.
- 7.1.2 Unless otherwise indicated, the kitchen exhaust hood should be balanced using the sliding dampers in order to achieve an even airflow throughout the hood structure.
- 7.1.3 Both exhaust and supply systems should be running whilst calibrating the system. Also consider external factors such as windows and air-conditioning.

7.2 Exhaust Airflows

- 7.2.1 The exhaust airflow is to be balanced using sliding dampers. These are accessed by removing the light cover and sliding the handles. Note these are locked in place with an allen key bolt.
- 7.2.2 In order to calculate the effective airflow a pressure tap is located in the centre rear of each module and the manometer reading here should match the figure corresponding to your prescribed airflow with all filters installed.
- 7.2.3 If even airflow can not be achieved due to ducting connection or if more flow is required at module or end of the hood than can be achieved additional damping can be made using the loose part number. Further to this dividers separating the plenums of each module can be purchased from UE.

7.3 Supply Airflows

- 7.3.1 The supply airflow is to be balanced using the dampers found within the supply spigot. These are accessed by unscrewing and removing the light cover panel panel.
- 7.3.2 The air velocity (metre/second) for the hood perforated front face is to be taken as a timed average across the face and as individual measurements in a grid. Eight testing points are required for 1.5 linear metres of perforated panel.
- 7.3.3 The average air velocity for each module of perforated panel is to be added and to reach the total and compared with the required flow.
- 7.3.4 Alternatively airflow can be measured with a pitot tube through the front grill to each supply spigot to measure the pressure corresponding to the desired airflow.
- 7.3.4 UE recommends a maximum supply air value on the hood of less than 80% of the maximum exhaust air value for each module in most cases. An average face velocity of 1.5 m/s or less through the front and side perforated grills of the exhaust hood is also recommended.

8. FIRE PROTECTION

The UE OPTI hood is designed to give you the maximum fire resistance with proper operation and maintenance.

8.1 Pre Installed Systems

Hoods can come pre fitted with fire suppression systems. In this case please refer to the specific instructions supplied with this equipment before working around it.

Pre-installed systems need to be regularly checked and maintained to keep them functioning properly.

8.2 Third Party Systems

Fire extinguishing systems can be installed in the hood by third party suppliers. It is the installer's responsibility to make sure the final set up is compliant with the local standards.

Holes cut in the hoods should be flashed with stainless or chrome steel and be fully sealed air tight

Notes:-

- That the area at panel 7 and 9 contains two separated layers of stainless steel and continuous glass.
- Panels at 8, 14 and 9 are removable for cleaning and infrequent maintenance.
- Filter parts 15 and 16 are also frequently removed for cleaning.

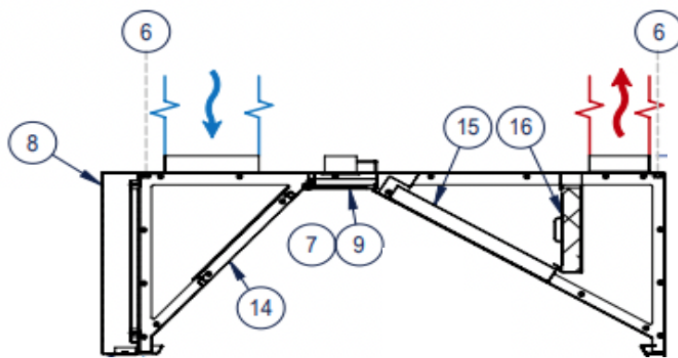


Figure 9
also see [Figure 4](#) for more detailed profile

9. OPERATION AND MAINTENANCE

9.1 Kitchen Exhaust Hood Cleaning

9.1.1 Cleaning and periodic maintenance are vital to ensure optimal performance of the UE kitchen exhaust hood and minimise service requirements.

9.1.2 In order to ensure efficient cleaning of the hood filtration equipment, TVB baffle and MCB honeycomb filters, it is recommended that spare sets of filters be purchased. Please contact your UE Agent to receive further information. Only certified UE parts are to be used within the exhaust hood.

9.1.3 It is recommended, in order to ensure optimal efficiency, that the honeycomb grease filters be periodically cleaned by a professional filter cleaning company. Day by day cleaning, for grease filters used under light cooking processes can however be done by cleaning the filters in a commercial kitchen dishwasher.

9.1.4 It is recommended that the inside and outside of the hood should be cleaned with hot water or a degreaser once a week. Never use caustic detergents or strongly acidic cleaners. Watch to make sure grease trays do not overflow or remove them. Rinse off cleaner residue and when the hood is dry, use a stainless steel cleaner to polish it. Baby oil can effectively prevent fingerprinting.

9.1.5 The exhaust and make-up chambers should be cleaned every three months using a non-caustic detergent. Access to the chambers is through the filter openings, by removing the access panels and make-up air grills at the front of the hood.

9.2 Troubleshooting

Note: All troubleshooting should be performed by qualified personnel and extreme caution should be used when working on the kitchen exhaust electrical circuits.

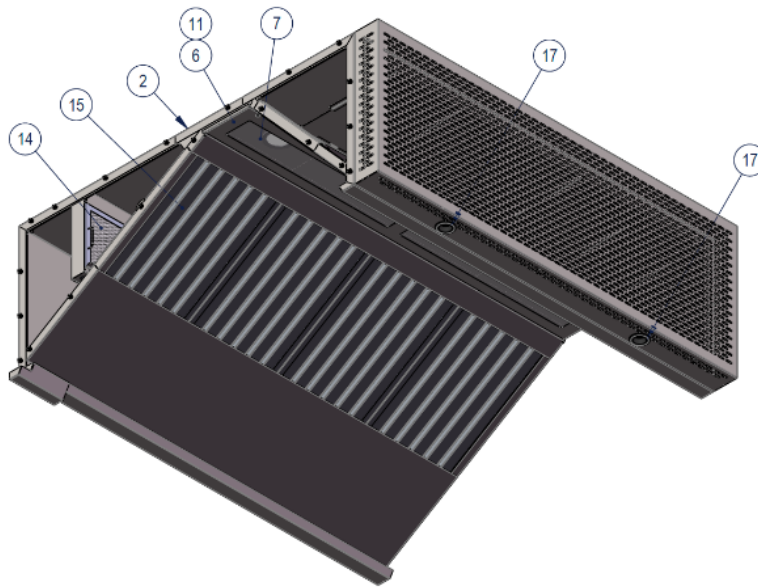


Figure 10

9.2.1 Kitchen Exhaust System Performance

9.2.1.1 The kitchen fills with heat and smoke

1. Check that the exhaust and supply air fans are operating. Often fans get dirty and old and this affects the performance of the system.
2. Are filters clean? Dirty filters will restrict the flow of air
3. Are windows or doors open or closed that are not normally? The outside wind direction can affect the air pressure in the room and the amount of air that exhausts.

9.2.1.2 Noise coming from the exhaust or supply duct

1. Try turning the exhaust fans off and on. They should wind up and down slowly and you can take note of when the sound starts and finishes.
2. If the sound is coming from the duct or fan you may need to clean or service this equipment.
3. If the sound appears to be coming from the hood and not the duct or the fans, try removing the filters this will change the pressure and velocity of air in the duct which may change the noise and give an indication of what is causing it.
Always have all appropriate filters installed in the hood when cooking.
4. Also try tightening bolts connecting and mounting the hood and holding the dampers in place. Damper fixings are found behind the light cover.
5. Contact your installer if issues persist.

9.2.2 Kitchen Exhaust Hood Fittings

9.2.2.1 An LED light is not operating

1. This may either be due to a connection problem with the power pack or due to a faulty LED strip that will need to be replaced.
2. Is there 240 V at the light junction box? Check the switch and power connection to the light junction box.
3. Is the LED light correctly connected to the power pack? Check the light connection with the power pack. If the light does not switch ON, the light is faulty and needs to be replaced.
4. Does the light work when connected to another power pack? (available on adjoining hood sections or for purchase separately)
5. Double check that the light does not work using another power supply if available.

9.2.2.2 The LED light is flickering or is partially or dim

1. This may either be due to a connection problem, a faulty power pack or due to faulty LEDs that will need to be replaced.
2. If all spots go out at once it is likely a power pack fault or issue with the power connection to this unit.
3. If one LED spot is out. Check the light connection with the power pack. If the light does not fully switch ON, the light is faulty and needs to be replaced.

9.2.2.3 Filters not fitting

1. High temperature differences can shrink and expand your hood and parts significantly. The hood is designed to get the best compromise between tight fit and easy install at different temperatures, however filters may be tight when the hood is cold and they are hot out of the dishwasher. You can also try different filters in each module as there may be some variation in filter sizes, due to manufacturing tolerances.
2. This could also mean your hood has moved since install, which is a structural concern for your building.

9.2.2.3 Hood leaking grease at the back wall

1. Check to make sure your grease cups are installed correctly and emptied.
2. You may also need to clean the inside of your hood.
3. If this does not resolve the issue you may need to get your duct cleaned as it may be leaking onto the hood from above.

9.2.2.4 Hood creating mist or condensation at the make up air vents

1. The difference between the air temperature and pressure of the air coming into the room and the air in the room is too wide.
2. If this occurs only on cold winter days, you may need preheating of your make-up air supply. Alternatively you can reduce the humidity both inside your premises and in the air supply.

9.2.2.5 Hood is making a rattling noise

1. Something may have been left in your hood or duct. Check it with a good clean.
2. If any bolts have fallen out these should be reinstalled with lock tight in a way that will assure this does not happen again, these may be structural, contact your installer.

10. WARRANTY AND CONTACTS

10.1 Contacts

10.1.1 For any questions concerning the information provided in these specifications, please contact your dedicated UE Agent. Contact details can be found on the UE website www.urbanenvironmental.co.

10.1.2 For any servicing and maintenance issues please contact your service provider who may not be UE. Try your mechanical engineer or HVAC installer, servicer or vat service company to provide hood, duct and fan servicing.

10.2 Quotation Validity

10.2.1 The validity of the UE quotation is limited to the items, drawings and specifications which are referenced (hereby called "the product"). Any modifications to the product will result in a quotation revision.

10.2.2 The validity of the prices quoted are current for 60 days from the date of quotation if not stated otherwise. A quotation extension may be accorded on request.

10.3 Sale Conditions

Purchased products cannot be returned for credit unless by written agreement from UE.

Once received, the condition of the product is the responsibility of the client.

10.4 Product Performance

10.4.1 UE selects in good faith and according to the information provided by others. The UE products are to be integrated into the overall mechanical design as per the UE defined specifications. In supplying equipment, UE cannot be held accountable for the overall performance of the mechanical system.

10.4.2 UE cannot be held accountable for product performance issues (for example excessive discharge of smoke, grease and odour) arising from the performance of the overall mechanical system including: duct design, fan selection, discharge point location, unknown cooking methods, and applied airflows other than the values specified by UE.

10.4.3 Filtration and odour mitigation products are specified based upon UE's in-depth industry experience and existing equipment efficiency tests.

10.4.4 The overall performance of the specified product can only be guaranteed if a preliminary assessment is carried out by a suitable UE consultant, taking into account cooking types and regimes, discharge point sensitivity, airflow amounts and the overall mechanical system design. This is notably the case with regards to odour discharge. UE underlines that additional filtration and odour mitigation units may be required should there be any perceived odour issues with the kitchen exhaust discharge during operations.

10.4.5 For kitchen exhaust hood purchases the UE product has been designed and certified according to international standards. The UE exhaust hood product specification and drawing sheet is to be approved and signed by the client, prior to

start of manufacturing. It is the responsibility of the client to ensure that all services to and from the hood are as required and that site access and installation constraints have been taken into account. The UE hood is suitable for its proposed application as given the requirements detailed on the drawing. It is the clients responsibility to ensure the set up is not changed or remains within EU parameters.

10.4.6 In order to assist installation, UE supply installation guidelines showing typical installation cases. These documents will be made available at the time of the purchase in electronic format. A hard copy is included together with the product. It is up to the Client to fully review the requirements of the product which in relation to deviations in the cookline configuration.

10.4.7 It is the responsibility of the Client to contact the local UE representative should they require any further information regarding performance, installation, commissioning and maintenance of the product. All UE products are to be installed following the locally applicable code and standards.

10.4.8 The commissioning of the product is to be carried out by a recognised mechanical contractor. All airflows should be as provided on the relevant specification sheets and documentation.

10.4.9 UE disclaims liability or responsibility for damage from fires and or explosions regardless of origin. It is the customer's responsibility to ensure that appropriate fire protection systems or explosion venting are installed following the requirements of local applicable codes and standards.

10.5 Maintenance and Cleaning

10.5.1 It is the client's responsibility to ensure that service access to UE equipment is per application codes, OHS regulations and UE specifications for the purpose of cleaning and servicing.

10.5.2 Cleaning and periodic maintenance is vital to ensure optimal performance of the UE exhaust and filtration equipment and to minimise service and replacement part requirements.

10.5.3 In order to ensure efficient cleaning of the filtration equipment (baffle filters, honeycomb filters, mesh filters, electrostatic cells), it is recommended that spare sets of filters and cells be purchased.

10.5.4 UE can provide details of maintenance companies to organise the maintenance and cleaning of filtration equipment. Please contact your local UE representative to receive further information

10.6 Transport and Delivery

10.6.1 Delivery of the product will be done to the address as indicated by the client and by means of the agreed logistics.

10.6.2 Unless otherwise mentioned, cost of delivery including freight and any additional handling costs (including crating), are not included in the quoted price.

10.6.3 Unless the client requests otherwise the delivery will be to the closest suitable access point, at which point the handover of the product is completed. The client is responsible for unloading of the product from the delivery vehicle, which may require specific lifting equipment. The client is responsible to refer to the UE product specifications concerning dimension and weight of the products and to contact their local representative should they require any further information.

10.6.4 The products are to be inspected upon arrival and, if applicable, compared with the packing list provided. If any damage or shortage is noted then the consignment note must be endorsed, the driver advised accordingly and UE immediately notified. Photographic evidence should be provided so that a claim can be filed.

10.6.5 Alternatively, the client can organise to have the product picked up at an UE manufacturing site or warehouse, at which point handover is completed.

10.7 Warranty

10.7.1 UE provides a limited warranty for metropolitan areas in the country of sale by the local UE distributor.

10.7.2 To validate the warranty, the product should be installed, maintained and cleaned as per UE specifications and requirements and the owner should sign an agreement with an UE accredited maintenance company and purchase spare filters and consumables for cleaning purposes. A maintenance logbook should be kept on the premises.

10.7.3 All new products are warranted to be free from defects and/or faulty workmanship for a period of one (1) year. The warranty period will start from the date of commissioning.

10.7.4 UE warrants that, during the Warranty Period, the product will, with normal use and with correct servicing, be free from faulty parts and defects arising during the Warranty Period.

10.7.5 If a client is not able to prove that the equipment has been cleaned and maintained as per UE requirements, this may be cause to void the warranty.

10.8.6 A replacement part is warranted to be free from defects and/or faulty workmanship for a period of ninety (90) days from the date of its installation.

10.8.7 Any claim must be presented to the UE head office. No warranty repairs will be granted without UE written consent.

10.8.8 The above limited warranty does not apply to damage resulting from accident, alteration, misuse or if the serial number is removed or defaced.

8. HOOD SPECIFICATION SHEET

Figure 11

Insert hood specification sheet here following and any notes below